

Table 1 | Predictive gains and the cost of compression

Primary: next-season wRC+ · Secondary: WAR per 600 PA · Full sample: five forecast years, 1,706 transitions

Model inputs	wRC+ MAE	wRC+ gain [95% CI]	WAR/600 MAE	WAR/600 gain [95% CI]
Current outcome (calibrated)	19.98	+0.92 [+0.58, +1.26]	1.642	+0.031 [+0.005, +0.058]
Fixed representative statistics	19.81	+0.75 [+0.45, +1.06]	1.617	+0.007 [-0.016, +0.029]
Training-selected representatives	19.71	+0.66 [+0.39, +0.92]	1.611	+0.0004 [-0.017, +0.018]
B2V	19.06	reference	1.611	reference
All 22 inputs	18.82	-0.24 [-0.43, -0.05]	1.571	-0.040 [-0.061, -0.019]
Per-tool PCA	19.00	-0.06 [-0.15, +0.0005]	1.607	-0.003 [-0.009, +0.001]
Unrestricted PCA	19.11	+0.05 [-0.05, +0.15]	1.614	+0.003 [-0.004, +0.009]
History-based projection	19.21	+0.37 [+0.17, +0.57]	1.589	+0.017 [+0.002, +0.031]
Projection + B2V	18.84	reference	1.572	reference
Projection + fixed representatives	19.07	+0.23 [+0.09, +0.37]	1.577	+0.005 [-0.006, +0.015]
Projection + selected representatives	18.60	-0.24 [-0.43, -0.06]	1.570	-0.002 [-0.018, +0.014]
Projection + all 22 inputs	18.55	-0.29 [-0.51, -0.07]	1.546	-0.026 [-0.048, -0.006]

Gain = row MAE – reference MAE; positive favors B2V. Rows 1–7: B2V. Rows 8–12: projection + B2V.

Lower MAE is better. Paired player-cluster bootstrap CIs (2,000 resamples); intervals crossing zero do not establish equivalence.

Per-tool PCA: one component per tool. Unrestricted PCA: five across 22 inputs. History-based projections: recalibrated Marcel-style.

Fixed representatives: ISO, Contact%, BB%, BsR, Def. Selected: training-only CV among the fixed set and 17 within-tool replacements.

Selection is separate for each target, forecast origin and standalone/history context; it never uses the held-out forecast outcomes.

The history+B2V versus history+selected-representatives wRC+ difference is inconclusive without 2020-related transitions.